Lab 02

Introduction to OP-AMPs

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2.1 Objective

The Objectives of this Lab are:

- To operate op-amp in inverting, non-inverting configuration and compute voltage gains.
- To nullify dc offset voltage in op-amp and operate op-amp as a buffer.
- To measure the slew rate of op-amp.

Pre-Lab: Task 1

Q1. What is an OP-AMP? What are the major applications of OP-AMP? Draw its symbol.

Q2. Read the datasheet of LM 741[1], and label the pin numbers in Figure 2.1. What do you understand by V^+ *and* V^- pins? State your answer in a sentence.

Q3. What is the Recommended Operating supply voltage? Use datasheet.

Q4. What is meant by terms MIN, NOM and MAX in the datasheet, which one of these values is recommended for OP-AMP Operation?

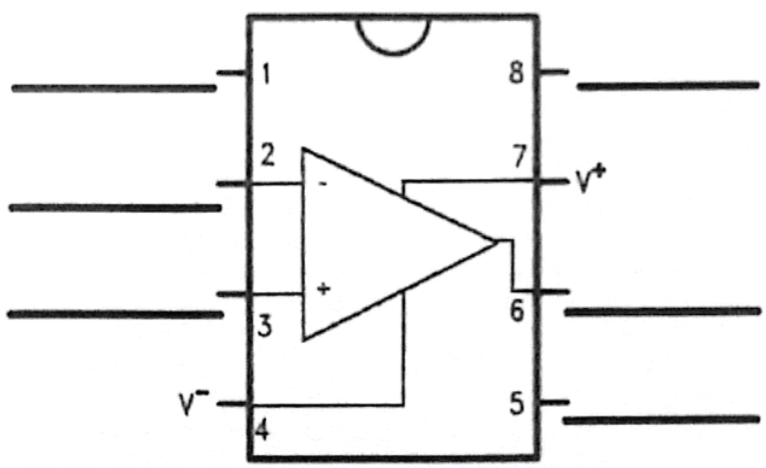


Figure 2.1: LM 741 Pinout Diagram

2.2 Introduction

The operational amplifier is a device that has all the properties required for DC amplification. It contains several stages and circuitry for temperature drift compensation. Although it could be made using discrete transistors it is usually an IC with all the components on a single silicon chip. This makes operational amplifiers available in large quantities at very low cost. The main problem of making a DC amplifier is to find a mechanism which ensures that the mean DC level of the output is

[1] Download Texas Instruments LM741 Datasheet from google.

1. Construct the circuit in Figure 2.6 and measure the output voltage record the value in Table 2.1

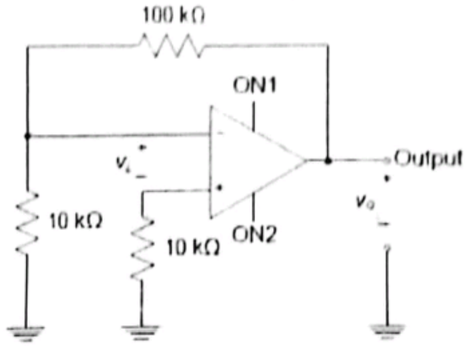


Figure 2.6

2. Construct the circuit in Figure 2.7 with potentiometer connected to the OP-AMP and set the potentiometer to such a point where V_{out} comes to be 0.00V. The schematic below shows two global connections, ON1 and ON2. In the physical circuit, these connections have to be realized via wires, i.e., the potentiometer must be connected to the offset null pins of the OPAMP, and the middle pin to the -V_{cc} of OPAMP. Record the output in Table 2.1.

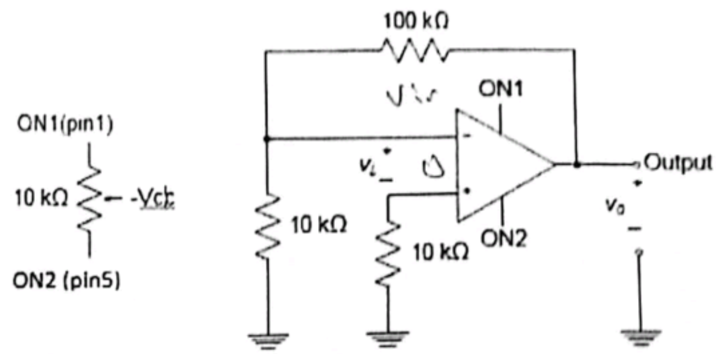


Figure 2.7 DC offset test circuit.

LR1 - 5
LR4 - 5

Table 2.1

V_{out} for OP-AMP without offset nullifying circuit	- 0.0133 V
V_{out} for OP-AMP after DC offset nullification	0 V

Task 3: Inverting Amplifier

1. Consider the circuit of Figure 2.4a using $R_1 = 20k\Omega$, $R_2 = 100k\Omega$. Calculate the voltage gain for the amplifier circuit and record in Table 2.2.
2. Construct the circuit of Figure 2.4a. Apply an input of $Vi = 0.1\ Vrms$, frequency = $10\ KHz$, $Va = 10V$. Using a scope, measure and record input and output voltage.
3. Compute and record voltage gain using measured values.

$\text{Gain} = -\frac{R_f}{R_i} = -\frac{100k}{20k}$

$\boxed{\text{Gain} = -5}$

4. Replace R_1 with a $100K\Omega$ resistor. Repeat steps 1 - 3.
5. Draw and label the waveforms in the box below for both the cases.

$$\text{Gain} = \frac{100k\Omega}{100k\Omega} = 1$$

Table 2.2: Inverting Amplifier

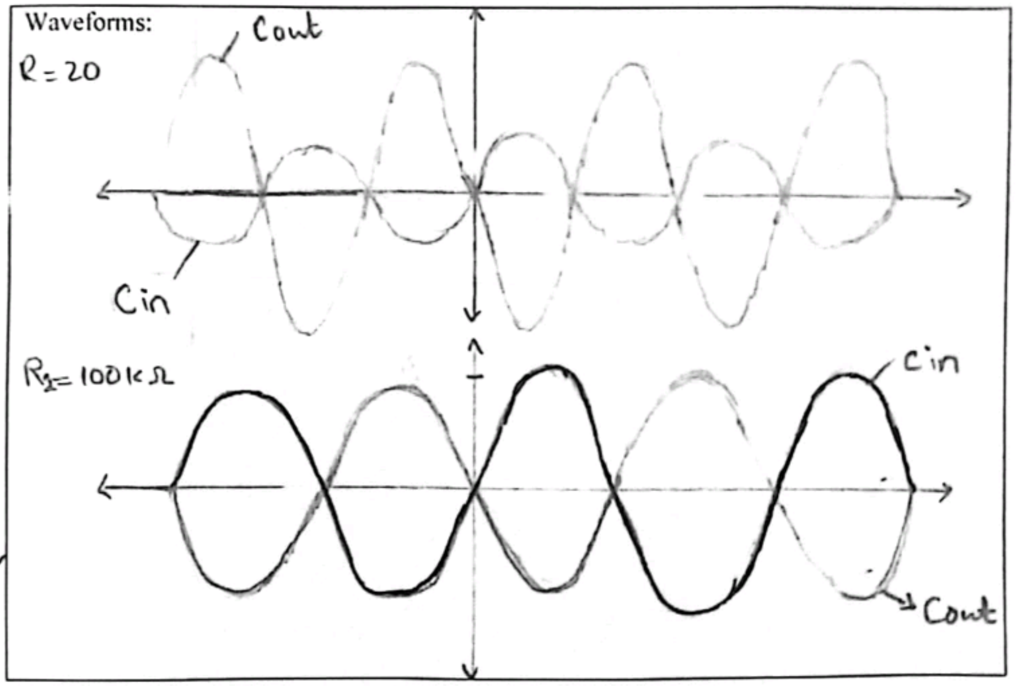
Resistances	Calculated Gain	Vin measured	Vo measured	Gain Observed
$R_1 = 20K\Omega$	5	103mV	515mV	5
$R_1 = 100K\Omega$	1	102mV	100mV	0.98

$$\frac{515mV}{103mV} = 5$$

$$\frac{100mV}{102mV} = 0.9$$

LR1 → 5

LR4 → 5



Task 4: Non-Inverting Amplifier

1. Consider the circuit of Figure 2.4b using $R_1 = 20k\Omega$, $R_2 = 100k\Omega$. Calculate the voltage gain for the amplifier circuit and record in Table 2.3
2. Construct the circuit of Figure 2.4b. Apply an input of $Vi = 0.1\ Vrms$, frequency = $10\ KHz$, $Va = 10V$. Using an Oscilloscope, measure and record input and output voltage.
3. Compute and record voltage gain using measured values.
4. Replace R_1 with a $100K\Omega$ resistor. Repeat steps 1 - 3.
5. Draw the waveforms for both the cases.

$$\text{Gain} = 1 + \frac{R_2}{R_1} = 1 + \frac{100{,}000}{20{,}000}$$

$$\boxed{\text{Gain} = 6}$$

$$\text{Gain} = 1 + \frac{R_2}{R_1} = 1 + \frac{100{,}000}{100{,}000}$$

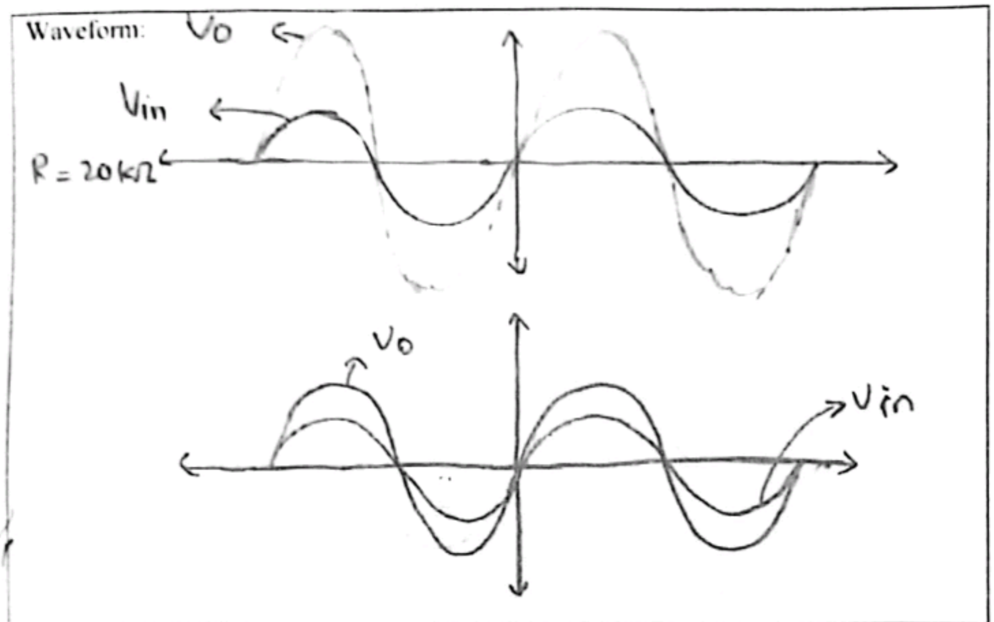
$$\boxed{\text{Gain} = 2}$$

Table 2.3: Non-Inverting Amplifier

Resistances	Calculated Gain	*Vin* measured	*Vo* measured	Gain Observed
$R_1 = 20K\Omega$	6	106mV	620mV	5.85
$R_1 = 100K\Omega$	2	105mV	203 mV	1.93

Gain = $\frac{620}{106}$ = 5.85

Gain = $\frac{203}{105}$ = 1.93



LR1 → 5

LR4 → 5

2.4 Slew Rate

The slew rate is defined as the maximum rate of output voltage change per unit time. It is denoted by the letter S. The slew rate helps us to identify the amplitude and maximum input frequency suitable to an operational amplifier (OP amp) such that the output is not significantly distorted.

The slew rate should be as high as possible to ensure the maximum undistorted output voltage swing. Slew rate is a critical factor in ensuring that an OP amp can deliver an output that is reliable to the input. Slew rate changes with the change in voltage gain. Therefore, it is generally specified at unity $(+1)$ gain condition.

A typically general-purpose device may have a slew rate of $10\ V/\mu S$. This means that when a large step input signal is applied to the input, the electronic device can provide an output of 10 *volts* in 1 *microsecond*.

The equation for the slew rate is given by:

$$Slew\ rate\ (S) = \frac{dVo}{dt}\Big|maximum \qquad Volts/\mu S$$

Task 5: Slew Rate

1. Build the circuit in Figure 2.9 on breadboard.
2. Connect Function generator at the input, set the function generator to output a square wave and set the input to 5V and frequency to 1KHz.
3. Connect the oscilloscope to the output of the op-amp circuit to measure the output waveform.
4. Use the oscilloscope to simultaneously capture the input and output waveforms. Ensure the time base and voltage scales are appropriately set to capture the entire waveform.
5. Identify the rising edge of the output waveform on the oscilloscope. Measure the time taken for the output waveform to rise from 10% to 90% of its final value (this represents the slew time) note the values in Table 2.4
6. Measure the $Vout$ and using measured $Vout$ calculate the 90% and 10% of the $Vout$. Fill the values in the Table 2.4.
7. Using Equation 3, calculate the slew rate of the OPAMP and record the value in Table 2.4. Show calculation in the table.

Table 2.4: Data Table

Output Voltage ($Vout$) (Measured)	5.12 V peak to peak
$t_{10\%}$ (Measured)	−2.60 us
$t_{90\%}$ (Measured)	2.20 us
$V_{o(10\%)}$ (Measured)	−2V
$V_{o(90\%)}$ (Measured)	2V
Slew Rate (S)	$S = \frac{dV_o}{dt} = \frac{2-(-2)}{2.60-(-2.20)}$ Slew Rate $= \frac{4.00\ V}{4.80\ us}$ $\boxed{\text{Slew rate} = 0.83\ V/us}$

2.5 Post-Lab

Task 6: DC Offset (PSpice)

1. Launch PSpice (Capture CIS). In Place Parts, search for "OP-AMP"; this is an ideal op-amp. Place it on schematic area with its both inputs connected to ground. Run time domain analysis and find V_{out}. Note your observations in Table 2.5.

2. Build the same circuit using non-ideal op-amp model. Place LM741 op-amp model with its both inputs shorted to ground. Run time domain analysis and find V_{out}. Note your observations in Table 2.5.

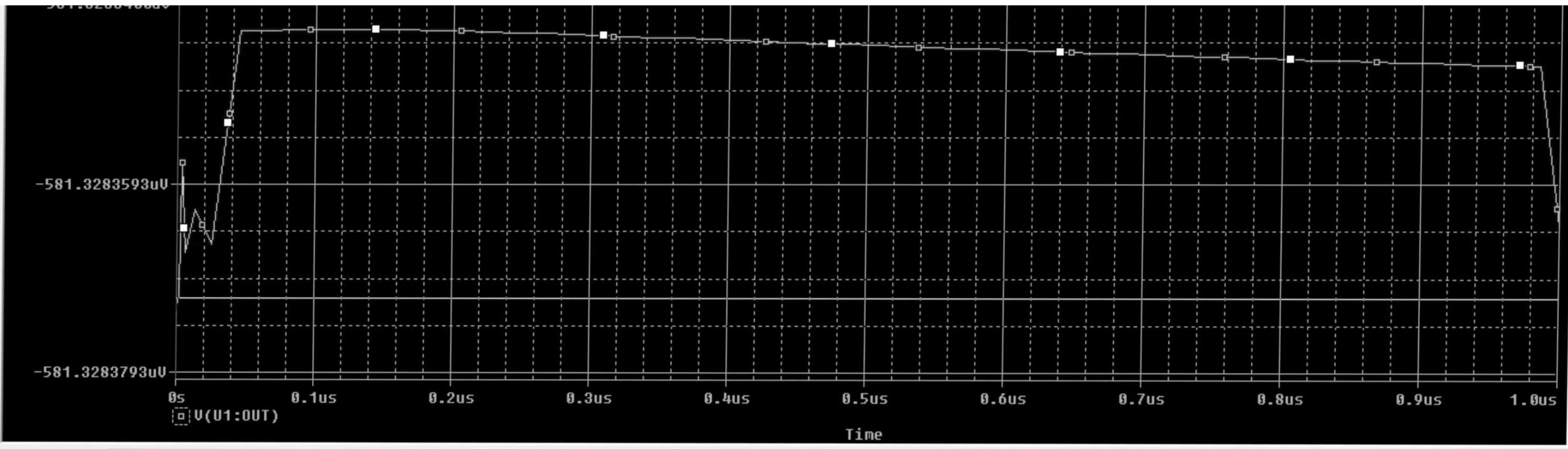
3. Compare the results obtained in task 1 and 6 and state the difference between ideal and non-ideal Op-amp.

Table 2.5

V_{out} for ideal OP-AMP	0V
V_{out} for non-ideal OP-AMP	-581-328uV

From the above observations, we can conclude that in an ideal OP-AMP, when two inputs are grounded, the voltage output would be 0V. This is because the difference ~~in~~ between the two inputs is 0.

While in the Non Ideal OP-AMP, some input offset voltage is there, Due to this gain regardless of the inputs being grounded, some voltage is amplified.



sim (active)

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		0.000		
	X Values	0.000	0.000	0.000		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1,2	V(U1:OUT)	-581.328u	-581.328u	0.000		0.000	0.000	-581.328u	-581.328u	-581.328u